TARUN SADARLA

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SUMMARY

Clinical AI & MLOps Engineer with a Master of Science in Artificial Intelligence (Biomedical Concentration, 4.0 GPA) and over 4 years of experience designing end-to-end machine learning systems. Specializes in Vision-Language Models (VLMs), computer vision architectures (CNNs, ViTs), and multimodal health records. Expert in architecting scalable pipelines, from multi-site clinical data harmonization and distributed computing (AWS, PySpark) to on-device edge inference and advanced neural network compression. Proven track record of deploying mission-critical deep learning applications with strict enterprise guardrails, robust CI/CD pipelines, and production-grade observability.

PROFESSIONAL EXPERIENCE

Molina Healthcare (Contract via Tekconnectedges) | Dallas, TX **September 2025 – Present**
AI / MLOps Engineer

- **Generative AI & RAG Pipelines**: Architected and deployed Vision-Language Models (VLMs) and Retrieval-Augmented Generation (RAG) pipelines using the AWS ecosystem to automate structured clinical reporting and adverse event detection from complex multi-modal health records.
- **Production MLOps & Kubernetes**: Built and scaled automated end-to-end MLOps pipelines (Data-to-Deployment) using Docker, Kubernetes (AWS EKS), and robust CI/CD gates, enabling zero-downtime, blue-green deployment of medical imaging models.
- **LLM Orchestration & Guardrails:** Designed agentic workflows using advanced orchestration frameworks and vector search engines, incorporating strict PII redaction and policy guardrails to enforce enterprise healthcare compliance.
- **System Observability:** Implemented real-time inference monitoring, data drift tracking, and model latency profiling across production deep learning workloads on AWS, significantly reducing post-deployment diagnostic cycles.

University of North Texas | Denton, TX **September 2024 – August 2025**
Graduate AI Research Assistant

- **Neural Network Compression & Fetal Ultrasound:** Invented a Hybrid Crossover pruning framework for structured CNN compression that synthesizes redundant feature maps rather than convolutional filters, utilizing gradient descent to reproduce the union of parent activations.
- **Temporal Segmentation:** Applied this pruning technique to a TemporalFetaSegNet (Residual U-Net) for automated fetal head circumference measurement from cine loops, reducing parameters by 43.7% (8.11M to 4.57M) while maintaining a 97.6% Dice score and 1.76mm MAE.
- **Synthetic Data Generation:** Developed Pseudo-LDDM v2, a four-stage physics simulator modeling Ornstein-Uhlenbeck probe drift, Rician speckle, and acoustic shadowing to generate realistic synthetic cine clips from static images.
- **Multi-Site MRI Classification (ASD):** Engineered a pure 5-layer CNN for multi-site brain MRI classification across 17 scanner sites (1,067 subjects), achieving a 0.994 AUC, 95.6% sensitivity, and 97.2% specificity. Implemented quality-weighted slice aggregation and 30-pass MC-Dropout to generate subject-level uncertainty estimates, routing high-variance predictions for human review.
- **Histopathology & Asymmetric Risk:** Developed a from-scratch histopathology cancer detection CNN (0.92 ROC-AUC) on H&E-stained tissue patches, deploying the model via FastAPI and Docker with a dedicated GradCAM endpoint. Designed a confidence-quartile subgroup analysis to flag the "gray zone" (p 0.3–0.7) for mandatory human review, deliberately optimizing for asymmetric clinical risk to prevent false negatives.
- **Edge-Deployed Sensor Fusion (WECARE):** Built an edge-deployed emergency detection system fusing heterogeneous ECG and IMU sensor streams via independent 1D CNNs, achieving a combined F1 score of 0.85. Engineered a late-fusion rule-based Emergency Alert Engine compiled with TorchScript, achieving sub-40ms on-device inference latency without cloud dependency.

Cotiviti Healthcare | Hyderabad, India **June 2022 – June 2024**
Data Analyst – Machine Learning Operations

- **Predictive Modeling:** Deployed supervised machine learning algorithms (XGBoost, Random Forests, and Ensemble methods) on high-volume clinical datasets to identify statistical patterns and predict risk anomalies.
- **Data Technologies & DWDM:** Developed scalable data warehousing and data mining workflows, applying PCA-based dimensionality reduction and feature engineering protocols to optimize tabular data schemas for downstream modeling.
- **Big Data Analytics:** Managed batch ETL pipeline workflows to ingest, clean, and validate unstructured clinical metadata, utilizing structured SQL queries and distributed computing principles on AWS EMR to stabilize ingestion throughput.
- **Performance Metrics:** Formulated comprehensive analytical dashboards to monitor statistical model drift, validating model precision, recall, and ROC-AUC curves against fluctuating historical data baselines.

Cotiviti Healthcare | Hyderabad, India **January 2022 – May 2022**
Data Science Intern

- **Clinical Data Preprocessing:** Spearheaded a targeted clinical data harmonization project, building preprocessing scripts to systematically normalize multi-source medical data and resolve structural site-specific schema inconsistencies.
- **Model Validation:** Assisted in validating diagnostic feature extraction modules by evaluating ensemble baseline performance, using structured Cross-Validation to ensure statistical stability across clinical subsets.
- **API Prototyping:** Collaborated with senior engineers to wrap predictive models into functional REST API microservices, validating data contract schemas and handling payload exceptions.

Tech Solutions | Remote, India **May 2021 – July 2021**
Machine Learning Intern

- **Foundational ML & EDA:** Completed structured technical training in Python, Exploratory Data Analysis (EDA), and core machine learning paradigms (classification, regression, and clustering).
- **Feature Engineering:** Built and trained predictive machine learning models on structured datasets, conducting feature selection, handling missing metrics, and performing hyperparameter tuning.
- **Evaluation Matrices:** Evaluated supervised learning systems using explicit statistical validation matrices, including Precision, Recall, F1-Score, and ROC-AUC curves.

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Computer Vision (CNNs, ViTs), Multi-Site Harmonization, Model Compression (Structured Pruning), Edge Inference
- **Generative AI & MLOps:** Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), LLM Orchestration, Agentic Workflows, Vector Search, CI/CD Pipelines, Blue-Green Deployments, Model Observability & Drift Tracking
- **Data Engineering & Big Data:** PySpark, Hadoop, Hive, AWS (S3, EMR, EC2, EKS), SQL, ETL/ELT Architecture, DWDM, Cassandra
- **Deployment & Tools:** Kubernetes, Docker, FastAPI, Flask, TorchScript, Git, REST APIs, Jupyter, Linux/Bash
- **Languages & Analysis:** Python, SQL, C, Pandas, NumPy, Statistical Validation (MC-Dropout, Cross-Validation), PCA, Feature Engineering

EDUCATION

University of North Texas | Denton, TX
M.S. in Artificial Intelligence, Concentration: Biomedical AI

Manipal Institute of Technology | Manipal, India
B.Tech. in Computer Science & Engineering, Minor: Data Analytics